



DARK TERRITORY

Abusing legacy railroad signaling systems



David Meléndez Cano
@TaiksonTexas

I+D & Embedded Software Engineer
Red Team en Innotec Security
Drone and Robot Builder
(Atropos & Interceptor Drones + Texas Ranger ROV)

Author of the book
"Hacking con Drones"



Gabriela García
@constrainterror

Security Software Developer
Hacker and hacking communities co-organizer
Coding & Cybersecurity instructor & mentor
Hardware Hacking Enthusiast



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In the railway sector, "dark territory" refers to a section of railway track that lacks remotely controlled railway signals or automatic block systems. In these areas, communication and coordination of train traffic rely exclusively on railway operators through radio communication and other non-automated methods, instead of using electronic signals or automated control systems to manage the safety and movement of trains.

The term has also gained popularity outside the railway context, especially in popular culture and information technology, to describe areas or sectors that lack supervision, regulation, or control, often symbolizing an environment of uncertainty, danger, or lack of control.



Railway Block: A block is a section of railway track in which normally no more than one train can be present at a time, in order to prevent a collision between two trains.



Block Section Between Stations: a section of railway track in which normally no more than one train can be present at a time, in order to prevent a collision between two trains



Block Section Between Signals: A block section between signals is a section of railway track in which normally no more than one train can be present at a time, in order to prevent a collision between two trains.



Axle Counter

A railway axle counter is a system of sensors installed on train tracks to detect and count the axles of the train cars.

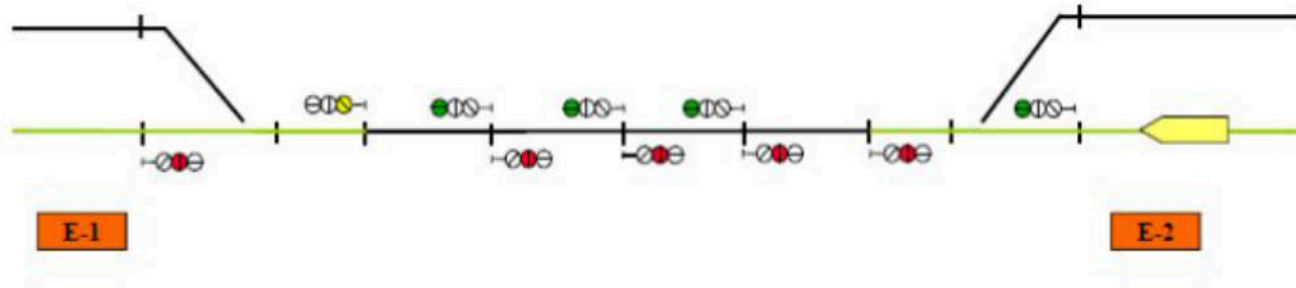


Track Blocking

A procedure by which the movement of trains on a track in a specific direction is authorized.



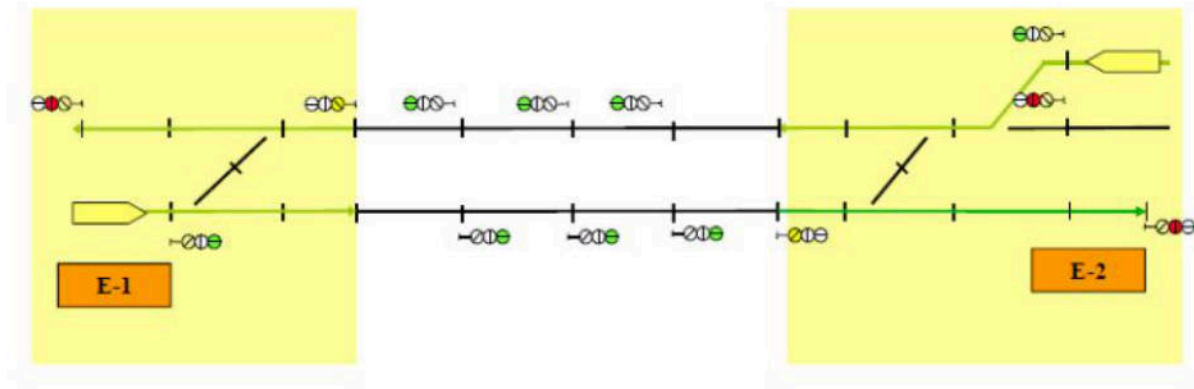
Electronic Track Blocking



Automatic Blocking on Single Track (B.A.U.)

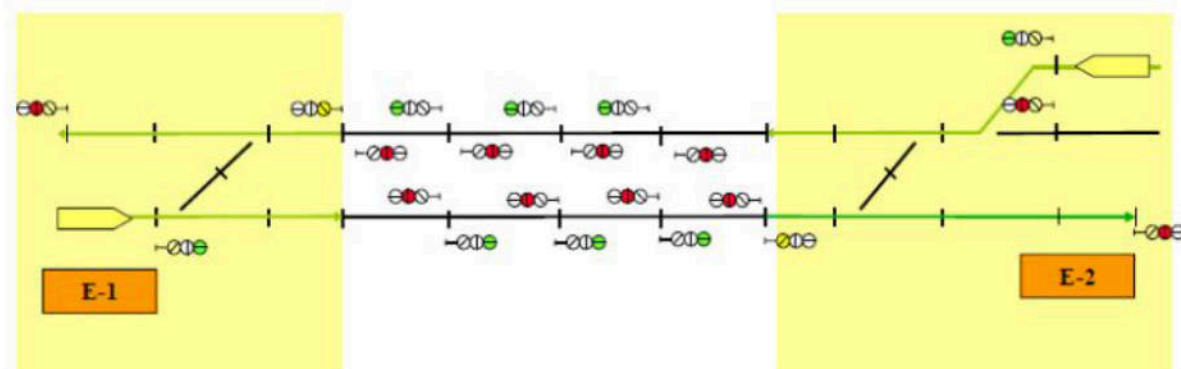


Electronic Track Blocking



Automatic Blocking on Double Track (B.A.D.)

Electronic Track Blocking



Bidirectional Automatic Blocking (B.A.B.)

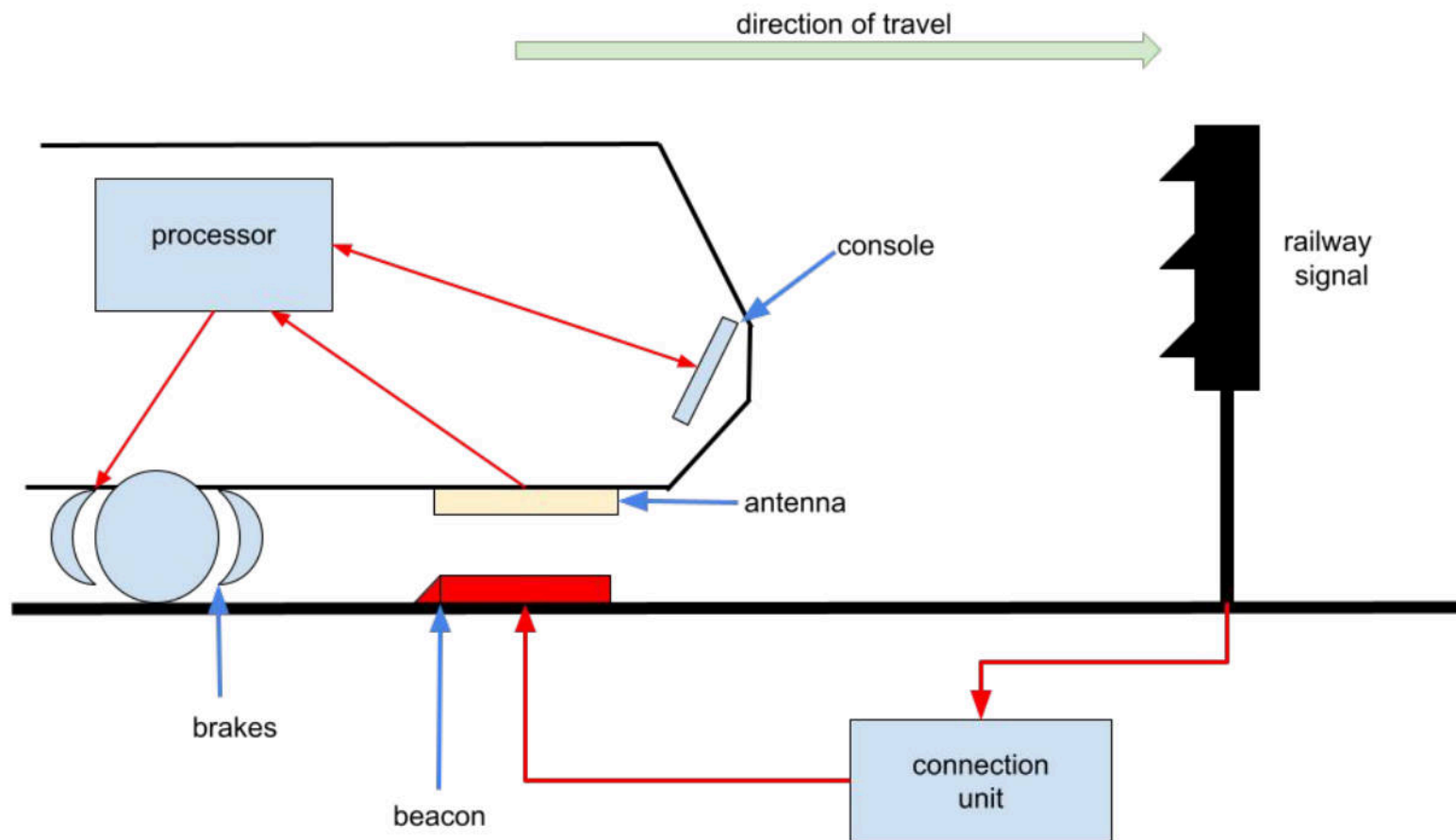
Centralized Traffic Control



ASFA (Anuncio de Señales y Frenado Automático) *Announcement of Signals and Automatic Braking*

This is the oldest support system for train circulation and is installed on almost the entire Spanish railway network. It is designed to reduce human errors. The system is based on a coil-capacitor circuit connected to the signal, which, depending on the signal aspect, transmits one frequency or another to the onboard equipment.





ASFA - Types

Beacons with Fixed Aspect - Example: Level Crossing Beacon

Beacons with Variable Aspect - Example: Signal Beacon

FASE 1	ANALÓGICO									
	L 3		L 2	L 1					L 7	L 8
	L 3	L 3 + REC PN	L 2	L 1 + RECA	L 1 + RECA+N	L 1 + REC V/A	L 1 + REC PN	L 1 + REC LTV	L 7	L 8
FASE 2	L 3	L 4	L 2	L 1	L 5	L 6	L 9	L 10 L 11	L 7	L 8

ASFA - Signal Frequencies

6.2. Frecuencias utilizadas en el sistema A.S.F.A.

Las frecuencias actualmente utilizadas son las siguientes:

L1—AMARILLO; VERDE-AMARILLO; AMARILLO DESTELLANTE	60.000 Hz.
L2—VERDE DESTELLANTE	64.020 Hz.
L3—VERDE	68.310 Hz.
L7—CONTROL DE VELOCIDAD	88.540 Hz.
L8—ROJO	95.500 Hz.

ANALÓGICO	L 3		L 2	L 1					L 7	L 8
FASE 1	L 3	L 3 + REC PN	L 2	L 1 + RECA	L 1 + RECA+N	L 1 + REC V/A	L 1 + REC PN	L 1 + REC LTV	L 7	L 8
FASE 2	L 3	L 4	L 2	L 1	L 5	L 6	L 9	L 10 L 11	L 7	L 8



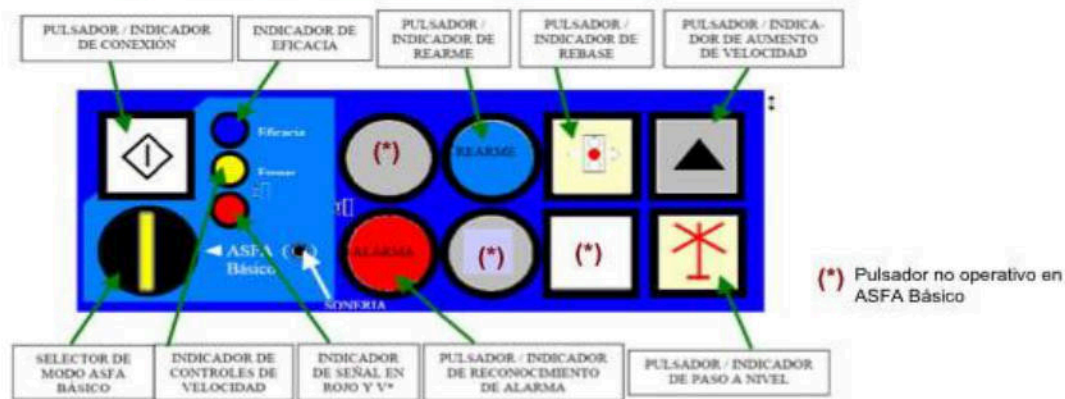
ASFA (Anuncio de Señales y Frenado Automático) - *Announcement of Signals and Automatic Braking*

Problem 1 - In Analog ASFA, the Same Aspect (e.g., L1 or L3) Indicates Multiple Signals

ANALÓGICO	L 3		L 2	L 1					L 7	L 8
FASE 1	L 3	L 3 + REC PN	L 2	L 1 + RECA	L 1 + RECA+N	L 1 + REC V/A	L 1 + REC PN	L 1 + REC LTV	L 7	L 8
FASE 2	L 3	L 4	L 2	L 1	L 5	L 6	L 9	L 10 L 11	L 7	L 8

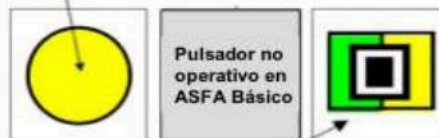
ASFA - What the Train Driver Sees

Panel ASFA Digital



Pulsadores adicionales de reconocimiento

Pulsador e indicación de reconocimiento de indicaciones L1



Pulsador e indicación de reconocimiento de vía libre condicional



L1: Anuncio de parada, anuncio de parada inmediata, preanuncio de parada, anuncio de precaución, paso a nivel desprotegido, anuncio de limitación temporal de velocidad.

ASFA - Lo que ve el maquinista





DARK TERRITORY



Uharte Arakil: 25 años del mayor desastre ferroviario de Navarra

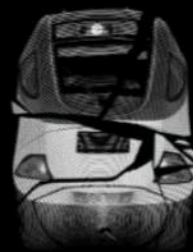
El intercity 'Miguel de Unamuno' descarriló en Uharte Arakil con 248 pasajeros, de los que 18 fallecieron. La tragedia marcó a una generación

UNA PELÍCULA DOCUMENTAL DE AITOR RUI

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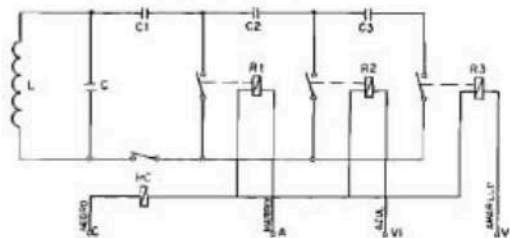


81 MUERTOS, MÁS DE 140 HERIDOS

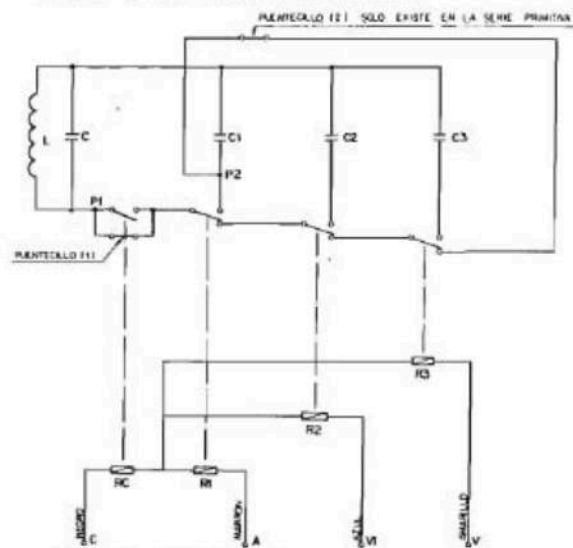
¿QUÉ HAY DETRÁS DEL ACCIDENTE FERROVIARIO
MÁS GRAVE DE LA DEMOCRACIA ESPAÑOLA?

CIRCUITOS TÍPICOS DE BALIZAS

Fig. 2



a) BALIZAS DE FONDO ROJO O CONTROL DE VELOCIDAD



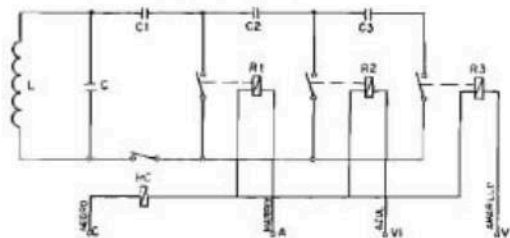
b) BALIZAS DE FONDO DOBLE

NOTA.- En las balizas de fondo doble que no son de la serie primitiva están unidos los puntos P1 y P2 con tan sólo el puenteado (1), coincidiendo el punto P1 con la posición de un contacto en reposo de RC y no pasando el camino hacia P2 por los contactos de R1, R2 y R3.

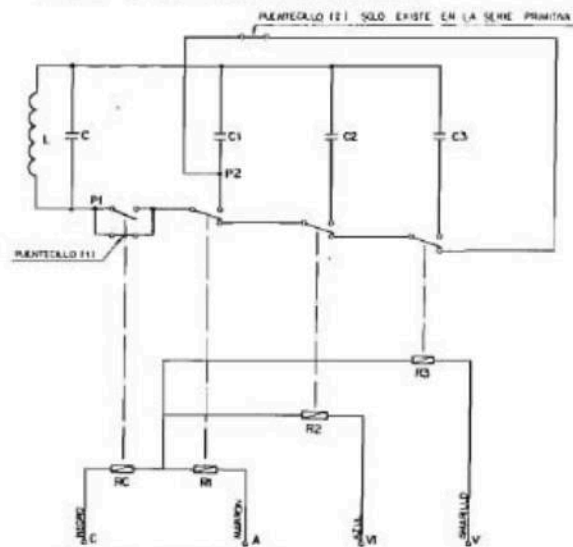


CIRCUITOS TÍPICOS DE BALIZAS

Fig. 2



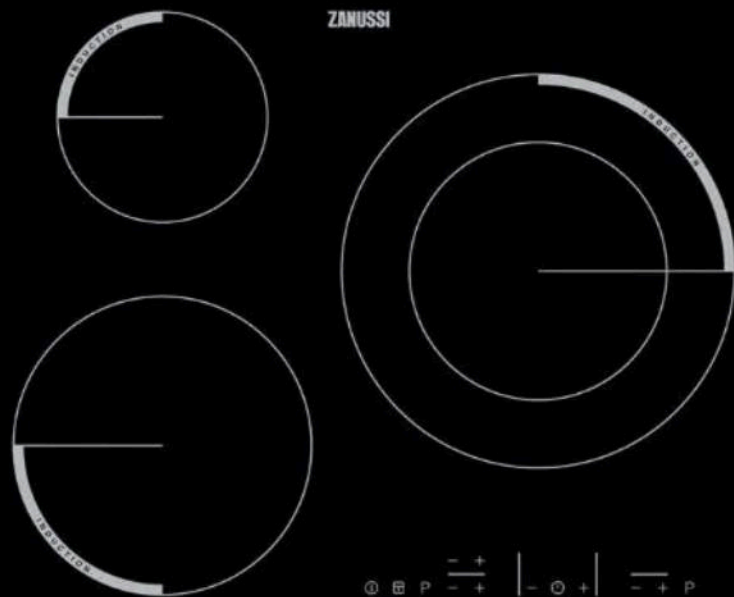
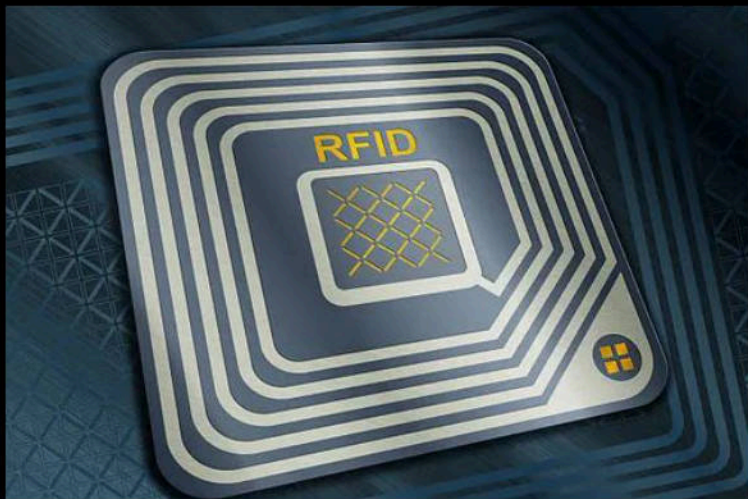
a) BALIZAS DE FONDO ROJO O CONTROL DE VELOCIDAD

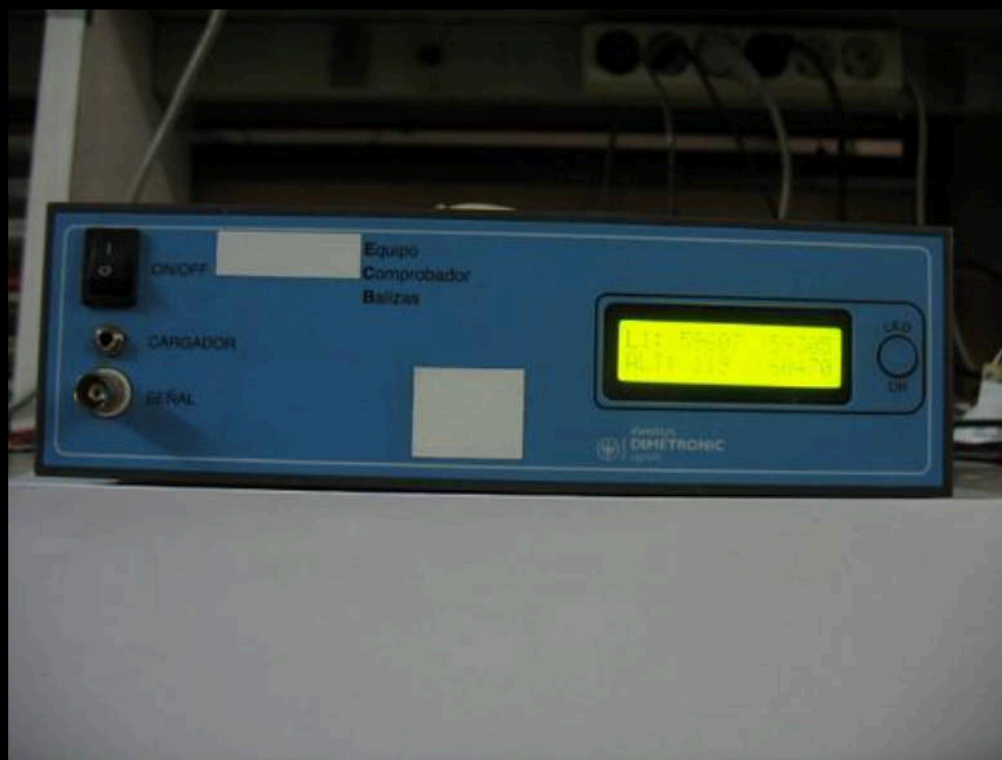


b) BALIZAS DE FONDO DOBLE

NOTA.- En las balizas de fondo doble que no son de la serie primitiva están unidos los puntos P1 y P2 con tan sólo el puenteado (1), coincidiendo el punto P1 con la posición de un contacto en reposo de RC y no pasando el camino hacia P2 por los contactos de R1, R2 y R3.





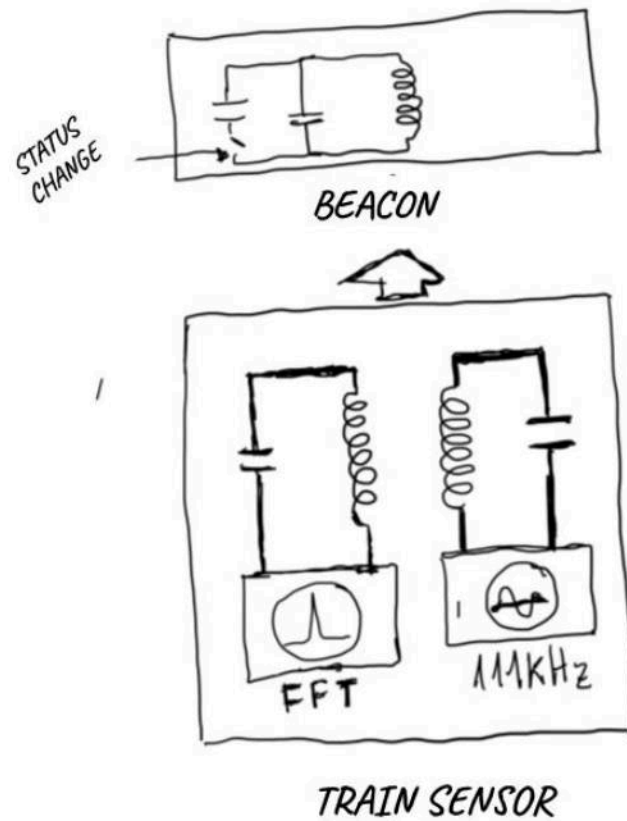


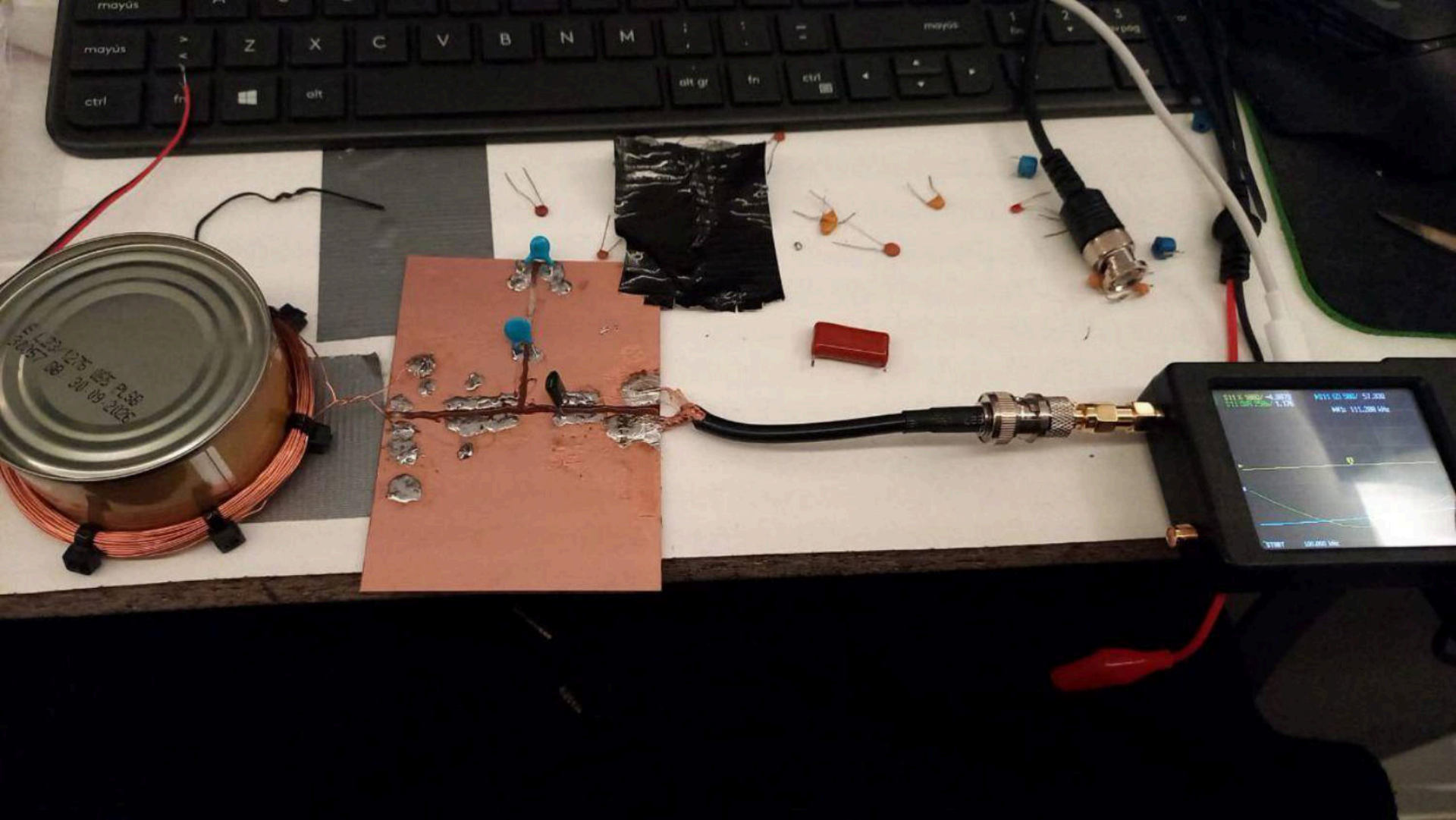
Possible Attacks:

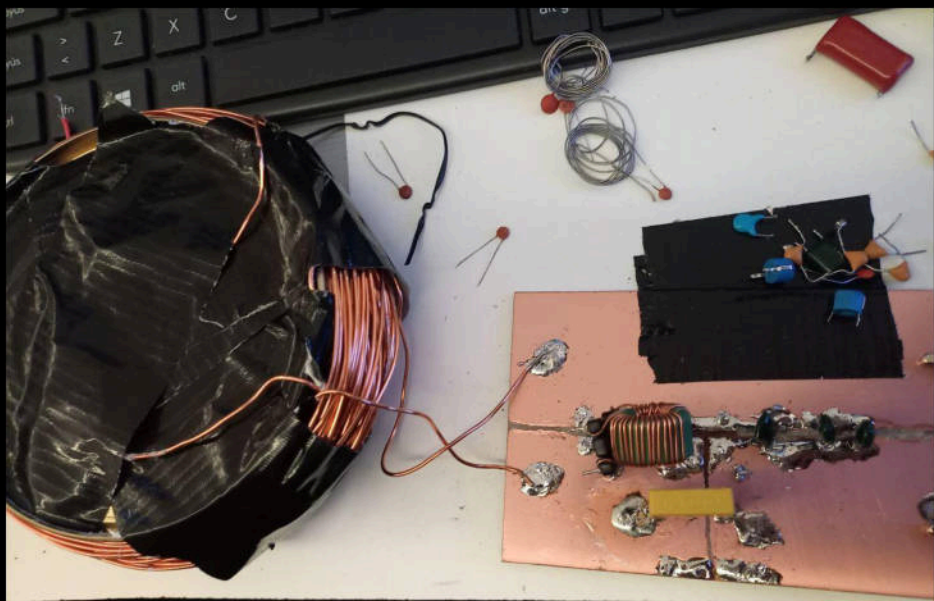
- Use a circuit with a coil and capacitor to induce the frequency we want the train to pick up to emulate a beacon.
- Replicate the beacon.

Requirements:

- Inductance of the coil
- Capacitance of the capacitor

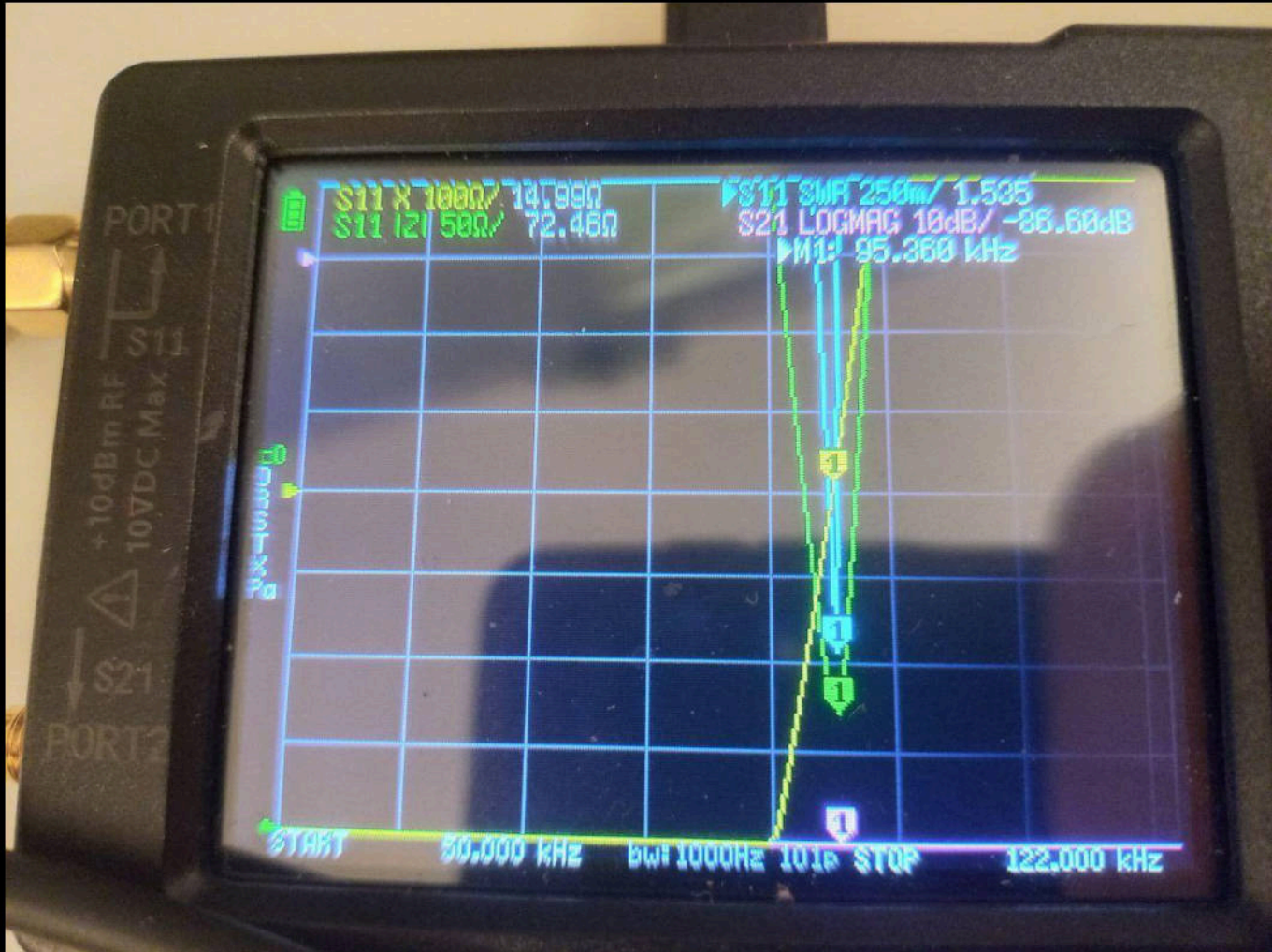






AND WHAT IF...





Thank You!

Acknowledgments:

Irene Cotillas

Pablo Trujillo

Matías

Pedro Candel (Saur0n)

@TaiksonTexas

@constrainterror